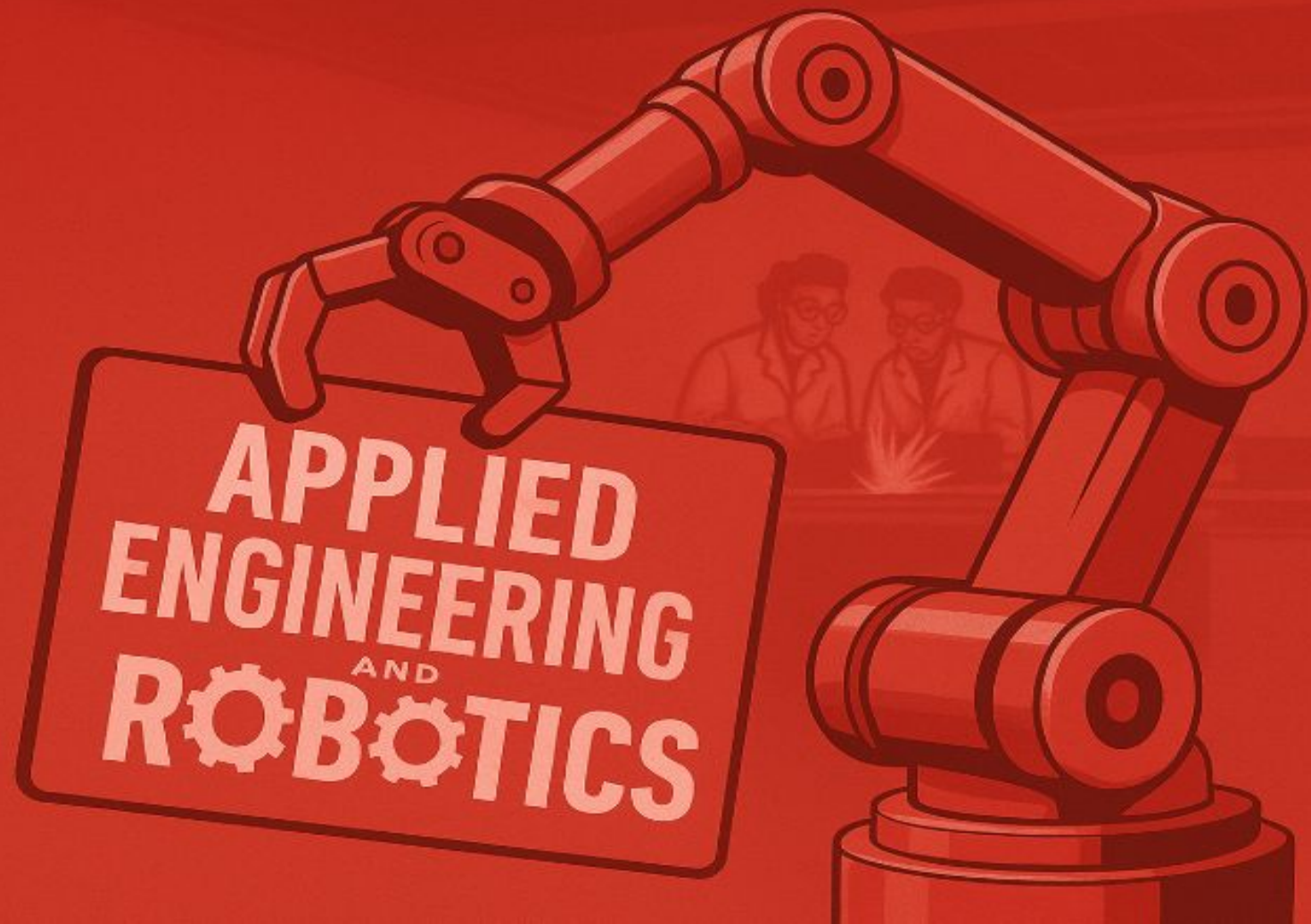


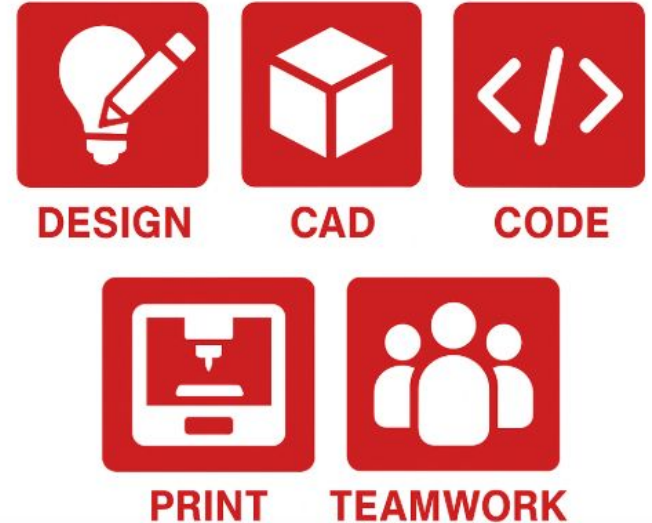


Welcome & Agenda

APPLIED
ENGINEERING
AND
ROBOTICS

What You'll Learn In This Course

- Master the Engineering Design Process
- Create functional CAD models & prototypes
- Program microcontrollers (Arduino/Robot brains)
- Program 3D printers, CNC Machine, and Plasma CNC
- Troubleshoot, and refine parts and designs
- Solve problems, build teamwork, project-management, and critical-thinking skills



Staying Safe in the Shop

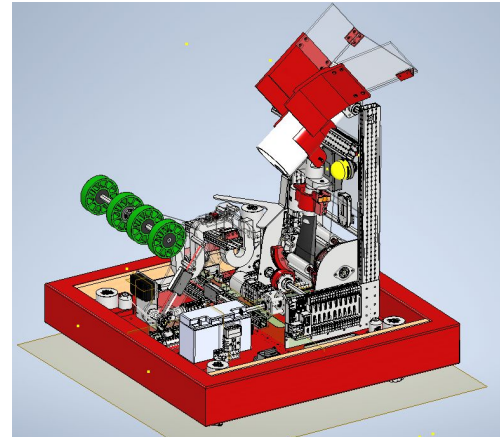
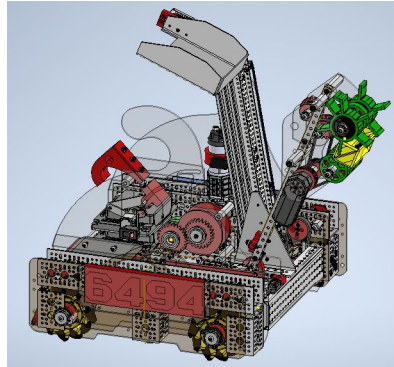
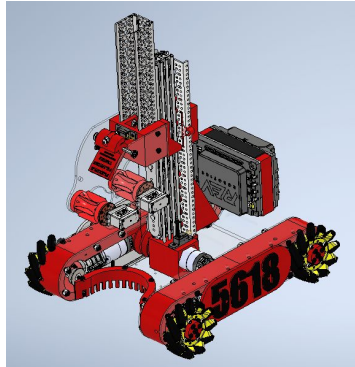
Top 5 Shop Rules

1. No horseplay
2. Always wear required PPE
3. No loose clothing or jewelry
4. Closed-toed shoes
5. Don't touch machines you're not trained on



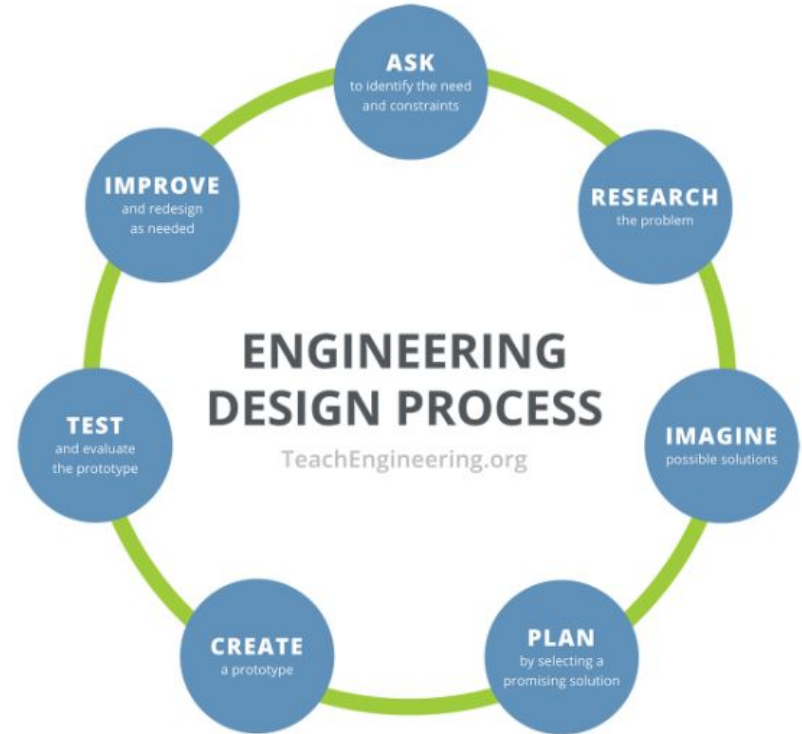
Explore Our Robotics Lab

- See past student robots & discuss design evolution
- Identify successes & “what would you change?” moments
- Note different mechanism types (drivetrains, arms, manipulators)



Steps of Engineering Design

1. Ask (Empathize, understand user needs)
 2. Research (clarify the problem)
 3. Imagine (brainstorm solutions)
 4. Plan (select a promising solution)
 5. Create (build a prototype)
 6. Test (gather feedback)
 7. Improve (refine iterations of your design)
- **Poll:** “Which step seems most critical to you?”



Ask – Empathize & Understand

Purpose: Put yourself in the user's shoes.

Key Activities:

- Interview stakeholders (e.g. customers, end users, anyone who will have something to say about the final product)
- Observe behaviors & pain points
- Create an empathy map (What they say, think, feel, do)

Prompt Question: “What really matters to our user?”



Research – Clarify the Problem

Purpose: Gather facts & define the scope.

Key Activities:

- Review incident reports / existing designs
- List requirements & constraints (size, cost, safety)
- Benchmark similar solutions

Prompt Question: “What do we know – and what don’t we know?”



Imagine – Brainstorm Solutions

Purpose: Generate as many ideas as possible.

Key Activities:

- Brainwriting / mind-mapping sessions
- Crazy-8 sketch exercise (8 sketches × 8 minutes)
- No-judgment “Yes, and...” culture

Prompt Question: “How might we...?”



Plan – Select a Promising Solution

Purpose: Narrow down to the best concept.

Key Activities:

- Evaluate ideas versus criteria (Pugh matrix)
- Identify materials, tools, timeline, and roles
- Sketch the key features & assembly steps

Prompt Question: “Which concept best meets our needs?”

	✓	✓	✗
1. 	✓	✓	✓
2. 	✓	✗	✗
3. 	✗	✗	✗
4. 	✗	✗	✗

Create – Build a Prototype

Purpose: Turn your plan into a tangible model.

Key Activities:

- Low-fidelity mock-ups (cardboard, foam, clay)
- CAD model export → 3D print / laser-cut parts
- Assemble and document build steps

Prompt Question: “Does it look and feel like we expected?”



Test – Gather Feedback

Purpose: See how your prototype performs.

Key Activities:

- Define test criteria (strength, fit, function)
- Conduct user trials / performance runs
- Record data, observations, and “failure modes”

Prompt Question: “What worked, and what didn’t?”



Improve – Refine & Iterate

Purpose: Use test insights to enhance your design.

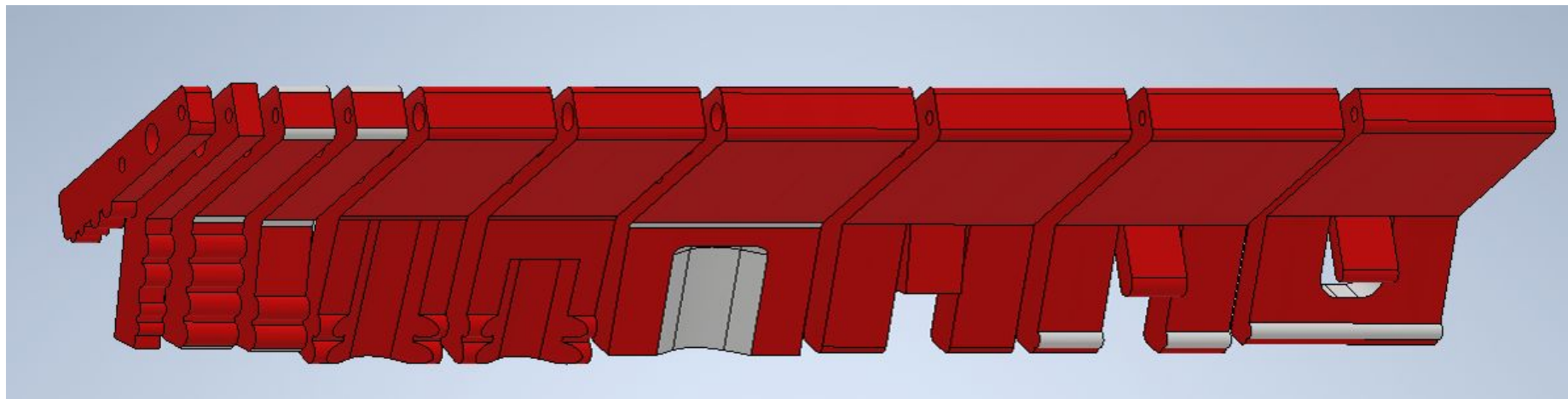
Key Activities:

- Update CAD models or prototype builds
- Retest the modified version
- Track version history & lessons learned

Prompt Question: “How can we make it even better?”



Improve – Refine & Iterate



Engineering Design Process

